

Solution of MGMT 670 Homework 1

Due 11:59 pm EST on September 3. Submit via Brightspace.

1. In smartphone battery testing, 12 phones were tested for battery life (in hours) under two different usage conditions: heavy gaming and normal usage. The following data were recorded:

Heavy Gaming: 4.8, 5.1, 5.3, 4.5, 4.2, 5.0, 4.7, 5.4, 4.9, 5.2, 4.6, 4.4

Normal Usage: 9.5, 10.1, 9.8, 9.2, 9.9, 10.3, 9.7, 10.0, 9.6, 9.4, 10.2, 9.3

- (a) Calculate the mean battery life under both conditions.

Solution: Mean (Heavy Gaming) = $\frac{58.1}{12} \approx 4.842$ hours

Mean (Normal Usage) = $\frac{117.0}{12} = 9.75$ hours

- (b) Calculate the median of the heavy gaming battery life.

Solution: Since there are 12 values, the median is the average of the 6th and 7th values:

$$\text{Median} = \frac{4.8 + 4.9}{2} = 4.85 \text{ hours}$$

- (c) Which usage condition has the higher sample standard deviation?

Solution: The sample standard deviation is

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}.$$

Heavy Gaming: $\bar{x} = 4.842$

$$s_{\text{HG}} \approx 0.375$$

Normal Usage: $\bar{x} = 9.75$

$$s_{\text{NU}} \approx 0.361$$

Heavy Gaming has a slightly higher sample standard deviation.

2. Suppose the distribution of weekly grocery spending is bell-shaped with a mean of 150 and a standard deviation of 20. Use the empirical rule to determine the percentage of data within each of the following ranges:

- (a) 110 to 190

Solution: Calculate the number of standard deviations from the mean:

$$110 = 150 - 2 \cdot 20 = \mu - 2\sigma, \quad 190 = 150 + 2 \cdot 20 = \mu + 2\sigma$$

By the empirical rule, approximately 95% of the data fall within 2 standard deviations of the mean.

95% of weekly grocery spending is between \$110 and \$190.

- (b) 130 to 170

Solution: Calculate the number of standard deviations from the mean:

$$130 = 150 - 1 \cdot 20 = \mu - \sigma, \quad 170 = 150 + 1 \cdot 20 = \mu + \sigma$$

By the empirical rule, approximately 68% of the data fall within 1 standard deviation of the mean.

68% of weekly grocery spending is between \$130 and \$170.

3. Consider the experiment of drawing a single card from a standard 52-card deck.

- (a) What is the probability of drawing a face card (jack, queen, or king)?

Solution: The probability is

$$P(\text{face card}) = \frac{\text{Number of face cards}}{\text{Total number of cards}} = \frac{12}{52} = \frac{3}{13} \approx 0.2308.$$

- (b) If two cards are drawn *without replacement*, what is the probability that at least one of them is a face card?

Solution: It is easier to use the complement rule:

$$P(\text{at least one face card}) = 1 - P(\text{no face cards}).$$

There are 40 non-face cards. The probability that both drawn cards are non-face cards is

$$P(\text{no face cards}) = \frac{40}{52} \cdot \frac{39}{51} = \frac{1560}{2652} \approx 0.5882.$$

Therefore,

$$P(\text{at least one face card}) = 1 - 0.5882 \approx 0.4118.$$

4. A cybersecurity firm monitors login attempts to detect suspicious activity. From past data, about 3% of all login attempts are from hackers. The system also found that:

- If attempt is from a hacker, there is a 90% chance it will trigger a security alert.
- If attempt is from a legitimate user, there is a 5% chance it will (falsely) trigger a security alert.

- (a) Given that a security alert was triggered, compute the probability that the login attempt was from a hacker.
- (b) The firm decides to automatically block login attempts if this posterior probability exceeds 50%. Should the system block an account whenever an alert is triggered? Explain.

Solution: Let H denote that the login attempt is from a hacker, and A denote that a security alert is triggered. We are given:

$$P(H) = 0.03, \quad P(A | H) = 0.9, \quad P(A | H^c) = 0.05$$

where H^c denotes a legitimate user.

- (a) **Posterior probability that a login attempt is from a hacker given an alert:**

By Bayes' theorem,

$$P(H | A) = \frac{P(A | H)P(H)}{P(A)}.$$

First, compute $P(A)$:

$$P(A) = P(A | H)P(H) + P(A | H^c)P(H^c) = (0.9)(0.03) + (0.05)(0.97) = 0.027 + 0.0485 = 0.0755$$

Therefore,

$$P(H | A) = \frac{0.027}{0.0755} \approx 0.3576 \approx 35.8\%.$$

- (b) **Decision about automatic blocking:**

The firm blocks login attempts only if $P(H | A) > 50\%$. Here, $P(H | A) \approx 35.8\% < 50\%$.

Conclusion: The system should *not* block an account automatically whenever an alert is triggered, because most alerts are false positives.

5. The file Stock_Price_NVDA_MSFT.xlsx contains daily closing prices for technology companies Nvidia (NVDA) and Microsoft (MSFT) in the past year (2024/8-2025/8).

- (a) Develop a scatter diagram with NVDA stock price on the horizontal axis and MSFT stock price on the vertical axis. What appears to be the relationship between these two stock prices?
- (b) Compute and interpret sample correlation coefficient. What does this value indicate about the relationship between these two stock prices.

Solution:

(a) NVDA prices are on the horizontal axis and MSFT prices on the vertical axis. The scatter plot below shows that as NVDA stock prices increase, MSFT stock prices also tend to increase, indicating a *positive relationship* between these two stocks.

Solution: (b) **Sample correlation coefficient:** Using minitab under “Stat → Basic Statistics → Correlation” tab, the correlation coefficient between the NVDA and MSFT price is

$$r \approx 0.87.$$

Interpretation: This indicates a strong positive linear relationship between NVDA and MSFT stock prices. When NVDA prices increase, MSFT prices also tend to increase.

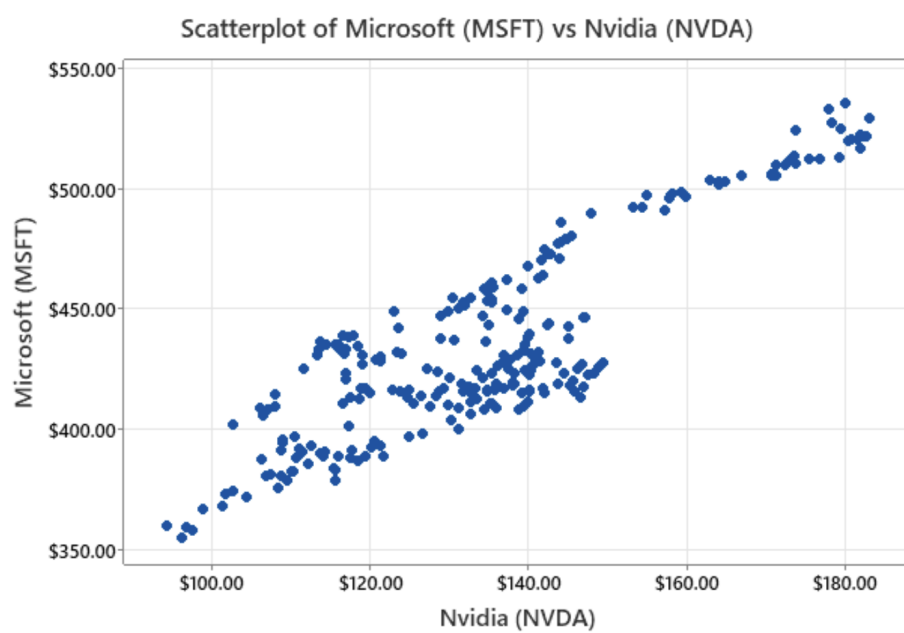


Figure 1: scatter diagram